
Software Requirements Specification

for

Mentor Caring System

Version 1.1 approved

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Revision History

Name	Date	Reason For Changes	Version
B09	2026.03.12	Initial approved version	1.0
B09	2026.03.20	Updated to align with Version 1.1 long description and supervisor feedback	1.1

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the **Mentor Caring System (MCS)**. The purpose of this Software Requirements Specification (SRS) is to describe the systems functions, external interfaces, constraints, and overall behavior.

This document is intended for project stakeholders including developers, project managers, instructors, and future maintainers of the system. It serves as a reference for system design, implementation, and testing.

1.2 Document Conventions

This document follows standard Software Requirements Specification conventions. Section titles and numbering follow the structure defined in the SRS template. Important terms and system components are written with consistent naming throughout the document.

Diagrams such as the context model and use case diagrams are included to visually represent system structure and user interactions.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

1. **Developers**, who will design and implement the Mentor Caring System;
2. **Project team members**, who need a shared understanding of the system scope and requirements;
3. **Instructors and evaluators**, who will review whether the requirements are complete, clear, and consistent with the project description;
4. **Future maintainers**, who may extend or revise the system in later stages.

Readers who want a general understanding of the system should begin with Section 2: Overall Description.

Readers who are interested in system integration and external dependencies should focus on Section 4: External Interface Requirements.

Readers who want to understand quality constraints and operational expectations should read Section 5: Other Nonfunctional Requirements.

1.4 Project Scope

The **Mentor Caring System (MCS)** is a web-based information system designed to support the **BNBU Mentor Caring Programme**, which aims to help undergraduate students adapt to university life more effectively through structured mentoring and follow-up activities. The system is intended to provide a centralized platform for managing mentoring-related information, communication, and records for the different participants involved in the programme.

The system supports multiple types of users, including **Administrator**, **Faculty Consultant**, Mentor, Student, MCP Coordinator, and optionally **Supporting Staff**. It is intended to improve the efficiency and consistency of mentoring management by digitalizing major mentoring-related activities.

The main goals of the system are to:

- allocate mentors;
- report and maintain interview records;
- support appointment arrangement between mentors and students;
- generate summary reports;
- track relevant user and mentoring information.

To achieve these goals, the system supports key mentoring management activities carried out by different user roles, including administrators, faculty consultants, mentors, students, and MCP coordinators. It is expected to improve the efficiency, consistency, and traceability of mentoring work by integrating mentor allocation, record management, appointment support, information search, and communication functions into one system. The system also helps preserve historical mentoring records and supports the import of organizational and mentoring data, which is important for maintaining continuity when mentoring assignments or group membership change.

From the university's perspective, MCS supports the operational objectives of the BNBU Mentor Caring Programme by enabling more systematic management of mentoring activities, improving communication among participants, and making student support information easier to maintain and follow up. In this way, the system contributes to a more coordinated and scalable mentoring process within BNBU.

1.5 References

Design and Implementation of a Mentor Caring System, Project Long Description, Version 1.0, 2025–2026 Semester 2.

2. Overall Description

2.1 Product Perspective

The Mentor Caring System is a new information system developed for the BNBU Mentor Caring Programme. It is designed to support mentor caring activities for undergraduate

students, including mentor allocation, record management, appointment arrangement, and communication.

The system is not intended to replace an existing product. It operates within the university environment and relies on existing institutional resources, especially the university account database for login and user identification, as well as related student and staff information sources.

The system also interacts with external supporting components, such as Excel files for data import and the university email or communication system for user notification. The context model in this section shows the system boundary and its main external interfaces.

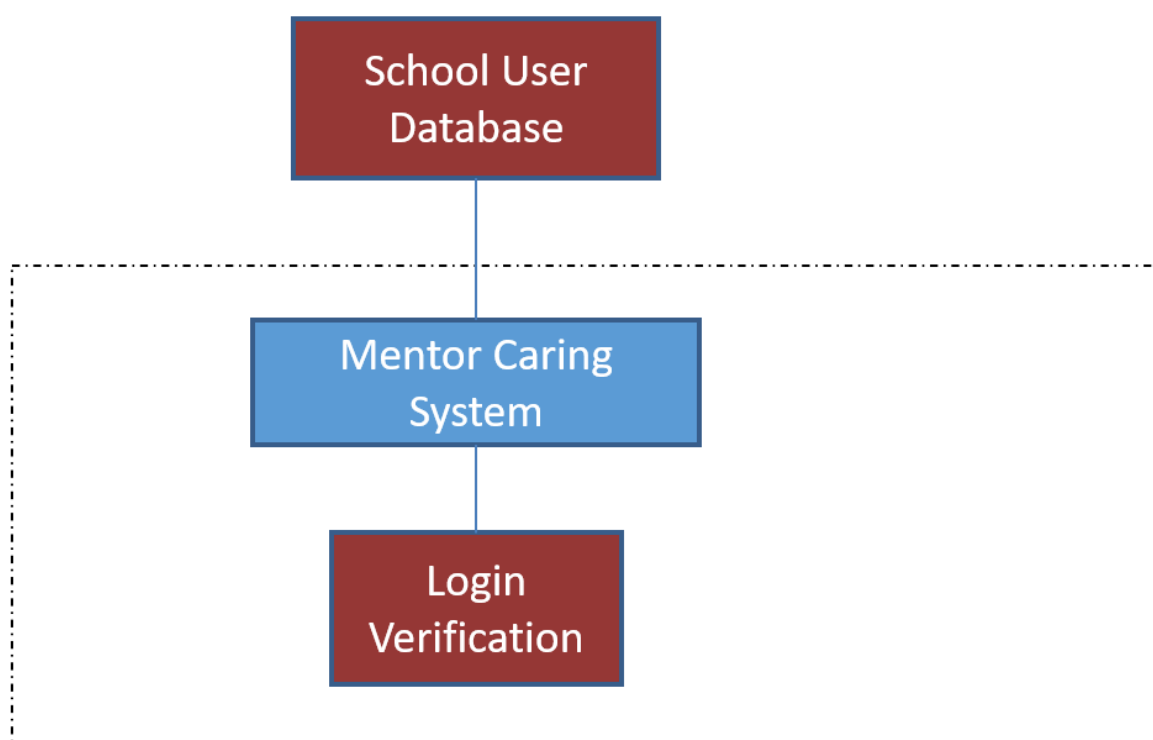


Figure 1: Context Model of the Mentor Caring System

2.2 Product Features

The Mentor Caring System supports the main activities of the BNBU Mentor Caring Programme. Its major features include role-based user access, mentor-group management, information search, interview record management, appointment arrangement, communication between users, special case forwarding, and record export. These functions help different users complete mentor caring tasks efficiently through one integrated system.

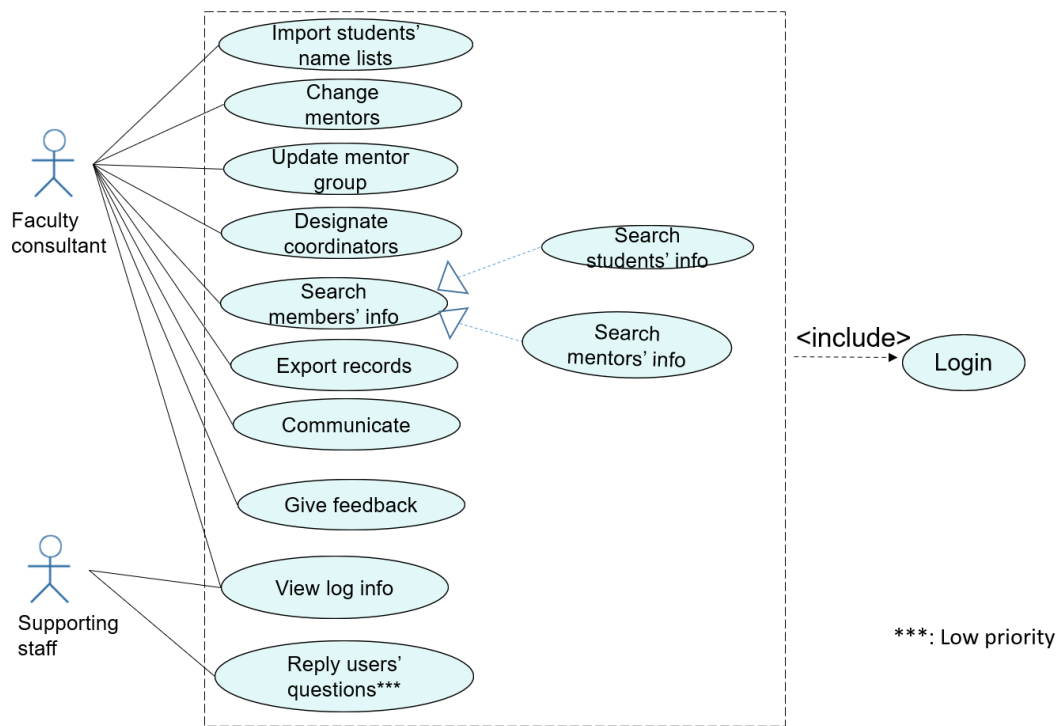


Figure 1

Figure 2: Use Case Diagram for Faculty Consultant and Supporting Staff

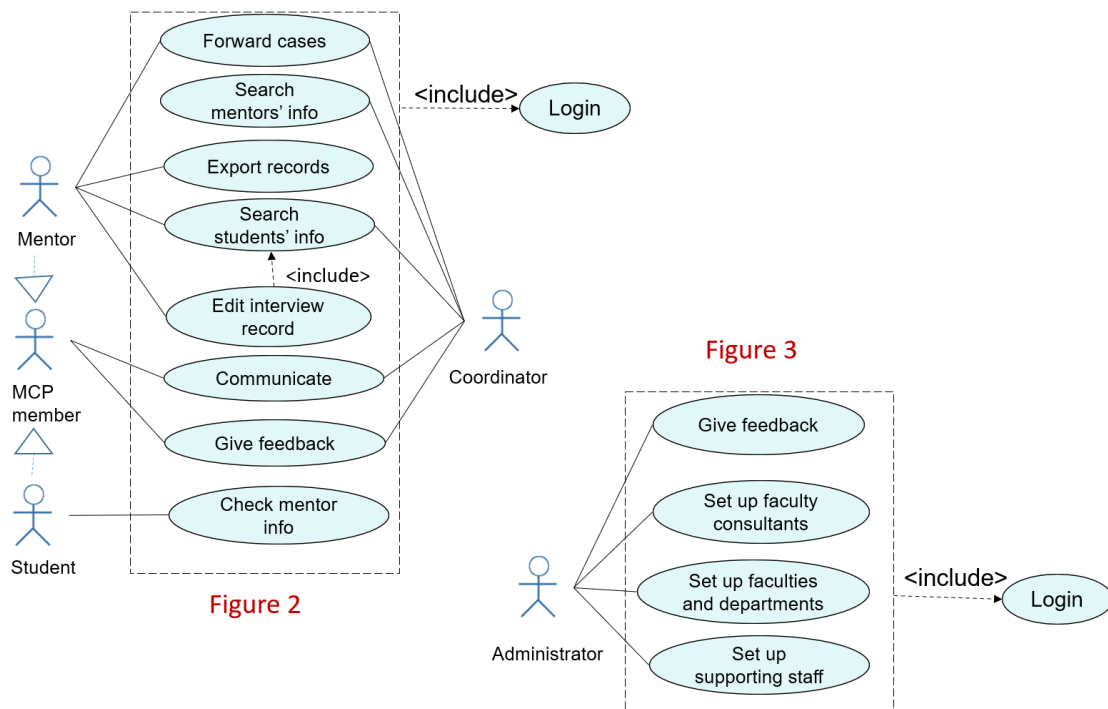


Figure 2

Figure 3

Figure 3: Use Case Diagram for Mentor, MCP Member, Student, Coordinator, and Administrator

2.2.1 Scenario for “Search Mentor Info”

Plan A – Basic Scenario

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.

6. The system displays the mentor's name associated with the group.
7. The system displays the record of one student in that group.

Plan B – Alternative Scenario: Group Not Found

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database.
6. The system cannot find the group information.
7. The system displays a message indicating that the Group ID is invalid and the use case ends.

Plan C – Alternative Scenario: No Student Record

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.
6. The system finds the mentor information associated with the group.
7. The system cannot find any student record in that group.
8. The system displays the mentor's name and a message indicating that no student record is available.

2.3 User Classes and Characteristics

The main user classes of the Mentor Caring System are administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.

- Administrator

The administrator is a privileged user responsible for overall system setup and management. Usually there is only one administrator in the university. This user designates and manages faculty consultants and supporting staff. The administrator has the highest management authority but is expected to use the system less frequently than faculty consultants, mentors, and students.

- Faculty Consultant

Faculty consultants are important management users at the faculty level. They are responsible for maintaining mentor allocation, managing mentoring groups, assigning MCP coordinators, searching information, and exporting records. They are frequent users, especially at the beginning of each academic year and whenever group allocation changes are needed.

- Mentor

Mentors are core operational users of the system. They mainly use the system to view students in their own groups, maintain interview records, communicate with students, arrange interviews, and handle special cases. They are frequent users and

generally have moderate computer literacy. Their access rights are limited to their own groups.

- Student

Students are end users of the system. Any current undergraduate student can log in to the system, but only students assigned to a mentoring group can continue to use its main functions. Students mainly use the system to check their mentors basic information and communicate with mentors. This user class is large in number and requires a simple and easy-to-use interface.

- MCP Coordinator

MCP coordinators are department-level users designated by faculty consultants. They are responsible for searching mentor and student information, handling special cases, and communicating with related users. In some cases, an MCP coordinator may also serve as a mentor. This user class has moderate to high access privileges.

- Supporting Staff

Supporting staff are optional users set up by the administrator. They mainly view student log information and may respond to user feedback about system usage. Since this function is optional, this user class is less important than the main operational user classes, and its usage frequency is expected to be relatively low.

2.4 Operating Environment

- The system depends on an existing school database that stores staff and student accounts, passwords, names, and roles.
- The system must support the import of Excel files for organization structure, mentor allocation, and MCP coordinator setup.
- The system must also support exporting information and interview records to Word files.
- The system must support the universitys email service to send email notification.

2.5 Design and Implementation Constraints

- The system shall use existing university accounts for user login rather than creating an independent account system.
- Import file format constraints: the system should strictly import file of Excel for setting faculties, departments, majors, student-mentor allocations, and MCP coordinators.
- Export file format constraints: the system should strictly export selected information and interview records to Word files.
- The system shall preserve historical student information even when mentors are changed or students are removed from groups.
- The system design shall be compatible with the existing school database and related university data sources.
- The system shall enforce role-based access restrictions for people with specific role could only access to specific functions and data scopes.

2.6 User Documentation

The following documentation will be delivered with the MCS:

- User Manual for general system use
- Administrator Guide for system administration
- Online Help integrated into the system
- Excel Template Files for required data import

2.7 Assumptions and Dependencies

- It is assumed that a university database containing valid staff and student accounts, passwords, names, and user roles is already available for authentication and identity management.
- It is assumed that faculty consultants will provide correct and properly formatted Excel files for setting up.
- It is assumed that student email addresses can be derived or obtained from student IDs.

3. System Features

3.1 Search Students Info

3.1.1 Description and Priority

Description:

This feature allows a Mentor or MCP Coordinator to search for a students information by entering the student ID. If the student record is found and the user is authorized to access it, the system shall display the students basic information and interview records. This feature serves as a prerequisite for Section 3.2 Edit Interview Record.

Priority: High

3.1.2 Stimulus/Response Sequences

Please see 3.2.2, Transition diagram is included in 3.2.2

3.1.3 Functional Requirements

REQ-1:

The system shall allow a Mentor or MCP Coordinator to search for student information by entering a student ID.

REQ-2:

The system shall validate the entered student ID before processing the search request.

REQ-3:

If the search is successful, the system shall display the students information, including student ID, group ID, mentor name, faculty/department/major, interview records, and status.

REQ-4:

The system shall allow a Mentor to access only the information of students in the mentors own mentoring groups.

REQ-5:

The system shall allow an MCP Coordinator to access only the information of students within the coordinators authorized department or scope.

REQ-6:

If no matching student record is found, the system shall display a clear error message.

REQ-7:

If the user is not authorized to access the requested student record, the system shall deny access and display a warning message.

REQ-8:

The system shall display the retrieved student information in read-only mode for this feature.

3.2 Edit Interview Record

3.2.1 Description and Priority

Actor: Mentor

Description:

This feature allows a mentor to edit a students interview record after the students information has been found through the use case Search Students Info (More information of Search Students Info, check 3.1). A mentor must search for students information first. The mentor could click Edit to add, modify, and delete interview. The mentor could modify and delete an existing record, and mentor could enter a new record if the record is empty. For any changes a mentor made, Save and Cancel button is provided for mentor to decide.

Priority: High

3.2.2 Stimulus/Response Sequences

The combined state transition diagram for Search Students Info (3.1) and Edit Interview Record (3.2) is shown in Figure 4.

3.2.2.1 Search Students Info Scenario

Actor: Mentor and Coordinator

Assumption:

The user has already logged into the Mentor Caring System with a valid account.

Basic Scenario:

1. The user selects the Search Students Info function.
2. The system displays a page for entering a student ID.
3. The user enters a valid student ID and submits the request.
4. The system validates the input and searches the database for the student record.
5. The system checks whether the user is authorized to access the student information.
6. If the student record is found and the user is authorized, the system displays the students information, including basic details and interview records.
7. The use case ends.

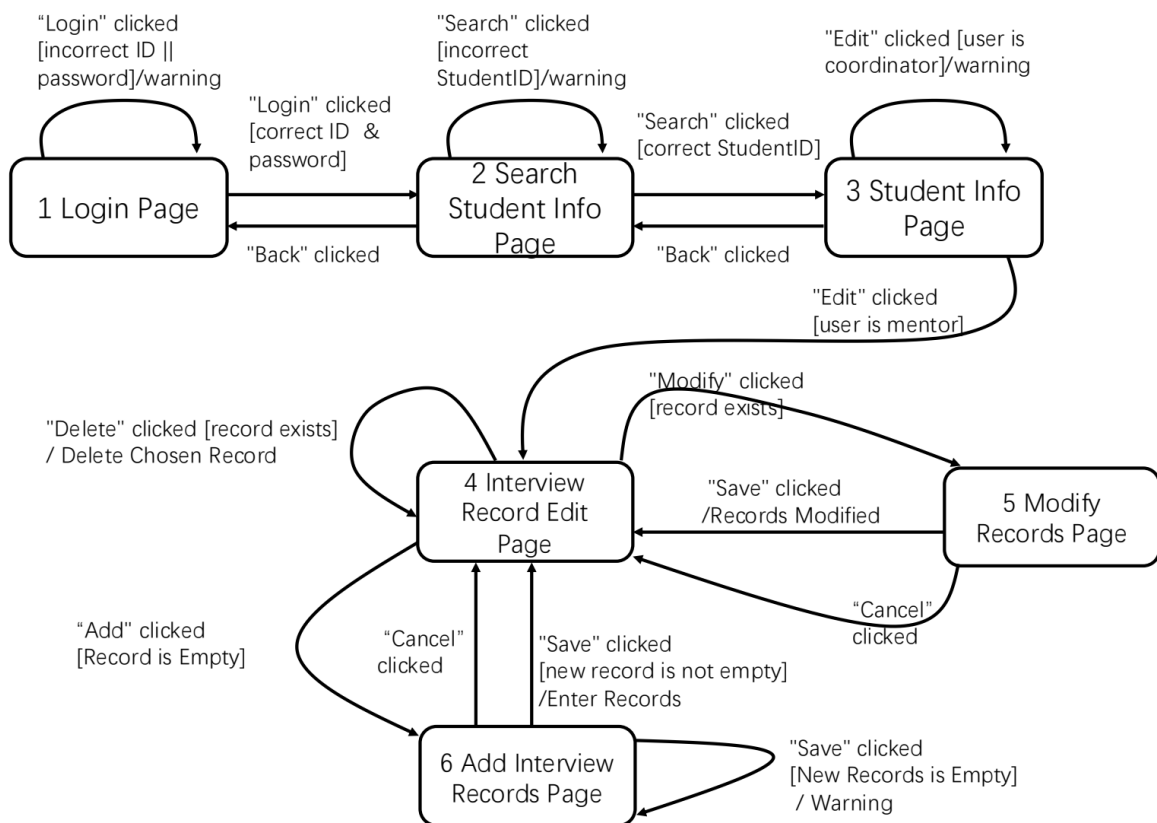


Figure 4: state diagram for Edit Interview Record

Alternative Scenario 1: Student Record Not Found

1. The user selects the Search Students Info function.
2. The system displays a page for entering a student ID.
3. The user enters a student ID and submits the request.
4. The system searches the database but cannot find a matching student record.
5. The system displays an error message indicating that no student record is found.
6. The use case ends.

Alternative Scenario 2: Unauthorized Access

1. The user selects the Search Students Info function.
2. The system displays a page for entering a student ID.
3. The user enters a student ID and submits the request.
4. The system finds the student record.
5. The system determines that the user is not authorized to access the record.
6. The system displays an access denied message.
7. The use case ends.

3.2.2.2 Edit Interview Record Scenario

Actor: Mentor

Basic Scenario:

1. The mentor logs in with his or her ID and Password.
 - The system will display warning if the ID or the password is incorrect.
2. The mentor views the students information page after searching by student ID.
 - The system will display warning if the student ID is incorrect.
3. The system displays the students basic information and current interview record, if any.
4. The mentor clicks Edit.
5. The system displays the interview record in editable form.
6. The mentor adds a new record, or modifies or deletes an existing record.
 - The mentor clicks Add to create a new record with the interview date and time, problem statement, interview summary, and follow-up actions.
 - The mentor clicks Modify of an existing record to modify the interview date and time, problem statement, interview summary, or follow-up actions as needed.
 - The mentor clicks Delete of a corresponding existing record.
7. The system displays a window with interview record, which could be added to new content or could be modified, and Save and Cancel after the mentor clicks Add and Modify.
8. The mentor clicks Save or Cancel to save or discard changes.
 - If the mentor clicks Save, the system stores the updated interview record and returns to the students information page with the latest content displayed.
 - If the mentor clicks Cancel, the system discards all unsaved changes and re-

turns to the students information page without updating the record.

9. The use case ends.

3.2.3 Functional Requirements

REQ-1:

The system shall allow only a mentor to edit interview records through this feature.

REQ-2:

The system shall require the students information to be displayed before the mentor can edit the students interview record.

REQ-3:

The system shall allow the mentor to add, modify, and delete an interview record for a student in the mentors own group.

REQ-4:

Each interview record shall contain the following items: interview date and time, problem statement, interview summary, and follow-up actions.

3.3 Search Mentor Info

3.3.1 Description and Priority

Actor: Faculty Consultant, Coordinator

Description:

This feature allows a user to search for mentor information by entering a Group ID. The system displays the mentors name associated with the group and shows student records in that group.

Priority: High

3.3.2 Stimulus/Response Sequences

The state diagram for the Search Mentor Info use case is shown in Figure 5.

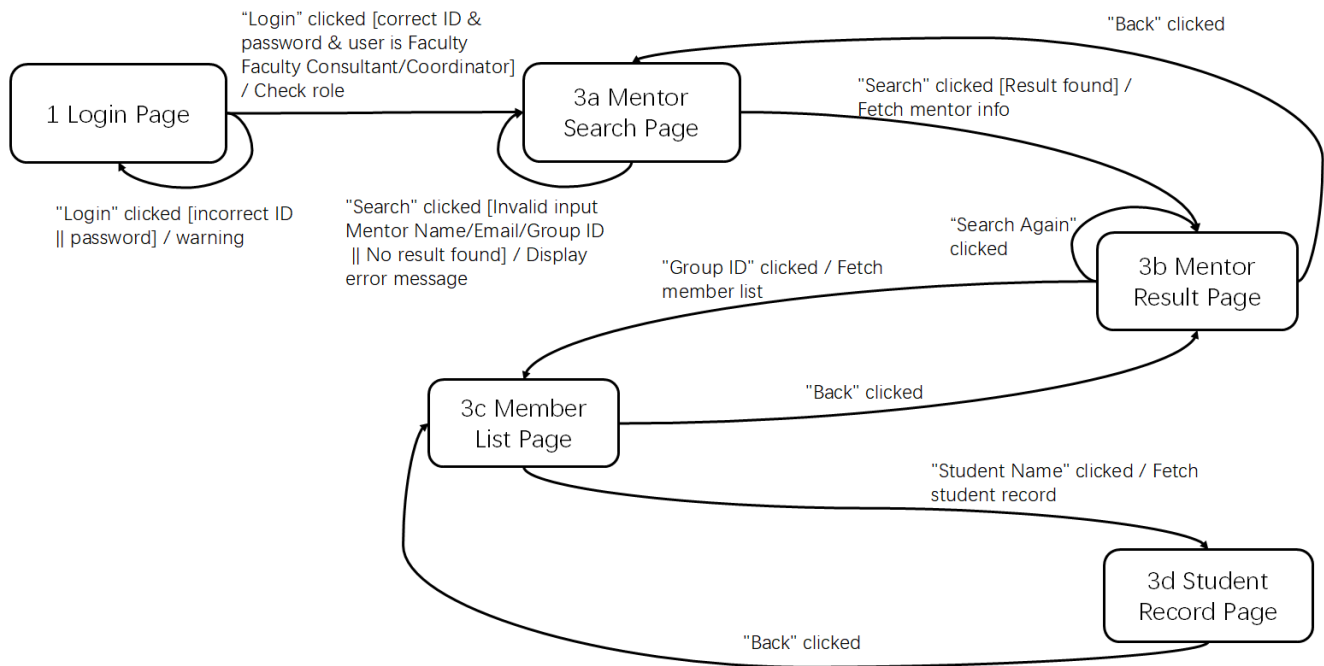


Figure 5: State Diagram for Search Mentor Info

Assumption: The Faculty Consultant or MCP Coordinator has already logged into the system and is authorized to access mentor information.

Basic Scenario:

1. The user is on the [3a Mentor Search Page].
2. The user enters a mentor name, email, or group ID and clicks the "Search" button.
3. The system validates the input, finds the record, and transitions to the [3b Mentor Result Page], displaying the mentor's name and the clickable Group ID.
4. The user clicks the Group ID button.
5. The system transitions to the [3c Member List Page], displaying the list of students in the group.
6. The user clicks a student's name from the list.
7. The system transitions to the [3d Student Record Page], displaying the student's detailed information.

Alternative Scenario 1: Invalid Search or No Result Found

1. The user is on the [3a Mentor Search Page].
2. The user enters an invalid name/email/ID or an ID that does not exist.
3. The user clicks "Search".
4. The system displays an error message, and the user remains on the [3a Mentor Search Page].

Alternative Scenario 2: Navigate Back and Search Again

1. From the [3b Mentor Result Page], the user clicks "Search Again".

2. The system navigates back to the [3a Mentor Search Page].
3. From the [3c Member List Page] or [3d Student Record Page], the user can click "Back" to return to the previous page in the navigation sequence as shown in the transition diagram.

3.3.3 Functional Requirements

REQ-1: The system shall allow users to search for mentor information using name, email, or group ID.

REQ-2: The system shall display mentor name and group ID in the search results.

REQ-3: The system shall allow users to view group members by selecting a group ID.

REQ-4: The system shall allow users to view student records by selecting a student name.

4. External Interface Requirements

4.1 User Interfaces

The user interface (UI) design for the system is provided in a separate document named: *UI_Diagram.pdf*

This document contains all UI screens corresponding to the state transition diagrams described in Section 3.

4.2 Hardware Interfaces

The **Mentor Caring System (MCS)** does not require direct control over specialized hardware. It relies on the underlying hardware of client devices and hosting servers to facilitate user interactions and data processing.

4.2.1 Supported Client Devices

Logical Characteristics: The system supports any device capable of running a modern web browser.

Physical Characteristics: Supported devices include desktop computers, laptops, tablets, and smartphones. The physical interfaces involve standard human-machine interface components: keyboards, mice, trackpads, and touchscreens for input, and standard displays for output.

Hardware Communication: Client devices connect to the server via standard physical network interface cards (NICs) supporting Ethernet (IEEE 802.3) or Wi-Fi (IEEE 802.11).

4.2.2 Server Hardware Interfaces

The **Mentor Caring System (MCS)** will be deployed on BNBU's server hardware. The server hardware must provide sufficient memory and CPU resources to handle concurrent database queries and file I/O operations, including parsing Excel files and generating Word documents.

4.2.3 Data and Control Interactions

User inputs are logically captured by the client browser and converted into HTTP/HTTPS requests sent to the server. The server hardware processes these requests and returns HTML/JSON data, which the client hardware renders onto the screen.

4.3 Software Interfaces

The **Mentor Caring System (MCS)** shall interact with external software components required for authentication, notification, and file processing.

4.3.1 School Database Interface

The system shall interface with the BNBU school database for user authentication. The database provides user account, password, name, and role information for staff and students. The system shall send authentication requests during login and receive validation results for access control.

4.3.2 Email Service Interface

The system shall interface with the university email service to send notification emails when a message is delivered in the system. Email notifications shall only inform the receiver that a message exists. Users shall still log in to the system to read and respond to messages.

4.3.3 Excel File Interface

The system shall support Excel file import for the following data:

- faculties, departments, and majors;
- student-mentor allocation data;
- MCP coordinator setup data.

The imported files shall follow the formats defined in the appendices. Imported student allocation data shall include student and mentor information required by the system.

4.3.4 Word File Interface

The system shall support Word file export for student interview records. Exported records shall follow the format defined in the appendix. The exported content shall include interview record details, including interview date and time, problem statement, interview

summary, and follow-up actions. Records of different majors shall be exported into different files.

4.4 Communications Interfaces

The **Mentor Caring System (MCS)** shall support communication between client devices, the application server, the school database, and the university email service.

4.4.1 Browser–Server Communication

Users shall access the system through a standard web browser. User operations, including login, search, record maintenance, messaging, appointment handling, import, and export, shall be transmitted between the client and the server through standard web communication mechanisms.

4.4.2 Database Communication

The system shall communicate with the school database during login to validate user identity and obtain role information. The communication shall support authentication for all system users.

4.4.3 Email Notification Communication

The system shall communicate with the university email service to send notification emails after messages are sent in the system. This communication shall support notifications for normal messages and other system messages that require user attention.

4.4.4 File Transfer Communication

The system shall support upload of Excel files from authorized users to the server and download of generated Word files from the server to authorized users. These communications shall support yearly allocation import, MCP coordinator setup import, and interview record export.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Mentor Caring System shall provide acceptable performance for normal operation.

5.1.1 Response Time

- User login authentication shall complete within 3 seconds under normal network conditions.
- Search queries for student, mentor, or log information shall return results within 3 seconds.

- A selected student record or mentoring group member list shall be displayed within 2 seconds.
- Sending an in-system message shall complete within 2 seconds, excluding external email delivery delay.

5.1.2 Import and Export Efficiency

- Import of valid Excel files for student–mentor allocation shall complete within 30 seconds for a normal yearly dataset.
- Import of valid Excel files for MCP coordinator setup shall complete within 30 seconds for a normal departmental dataset.
- Export of student or interview records to a Word file shall complete within 30 seconds for a normal dataset.
- Record filtering for export by academic year, department or major, mentor name, and student name shall complete without noticeable delay.

5.1.3 Concurrent Usage

- The system shall support at least 100 concurrent active users during normal semester operation.
- Operation logs shall be recorded in the background without interrupting normal user operations.
- Normal user operations shall remain available during yearly mentor allocation import.

5.2 Safety Requirements

N/A

5.3 Security Requirements

The system shall provide login verification, access control, privacy protection, and audit support for student records, interview records, messages, appointments, and operation logs.

5.3.1 Login verification

- All users shall be authenticated through valid university accounts and passwords from the school database.
- Authentication shall be required before accessing protected functions.
- Users shall log in to the system before reading messages, even if email notifications are received.

5.3.2 Access control

- The system shall enforce role-based access control.

- Faculty consultants shall access only information related to their own faculty.
- Mentors shall access only records of students in their own mentoring groups.
- Students shall access only their own mentoring information.
- MCP coordinators shall access only mentor and student information within their own department.
- Supporting staff shall access only functions assigned to supporting staff.

5.3.3 Privacy Protection

- Student records, interview records, message records, and log information shall be protected from unauthorized access.
- Email notifications shall not disclose sensitive message content.
- Only authorized users shall perform Excel import and Word export operations.

5.3.4 Auditability and Session Security

- The system shall record login time, logout time, and major user operations for audit purposes.
- Log information shall be protected from unauthorized modification or deletion.
- User sessions shall be invalidated after logout.

5.4 Software Quality Attributes

The system shall emphasize adaptability, reliability, usability, maintainability, correctness, and interoperability.

5.4.1 Adaptability

- The system shall support the addition of new faculties, departments, and majors without program code modification.
- Organizational structure information shall be configurable manually or through Excel import.
- Annual mentor allocation updates shall preserve existing student information.

5.4.2 Reliability

- Historical student information shall remain preserved after yearly mentor allocation import.
- Student information shall remain accessible after group removal or mentor reassignment.
- The system shall prevent silent data loss during import, export, update, and delete operations.
- The system shall achieve at least 99% availability during normal semester operation, excluding scheduled maintenance.

5.4.3 Usability

- Users shall see only functions appropriate to their roles.
- The system shall provide clear warning and error messages.
- The interface shall use consistent labels and navigation structures.

5.4.4 Maintainability

- The system shall adopt a modular design for user management, record management, messaging, import/export, and logging.
- Configuration data shall be separated from core business logic.
- System logs shall support debugging, auditing, and maintenance.

5.4.5 Correctness

- When duplicated data appear in an imported Excel file, existing valid information shall remain unchanged.
- Exported records shall match the selected academic year, department or major, mentor name, and student name.
- Access restrictions shall be enforced according to user role and organizational relationship.

5.4.6 Interoperability

- The system shall integrate with the school account database for authentication and role identification.
- The system shall support Excel import and Word export in the required formats.
- The system shall interoperate with an email service for message notifications.

6. Other Requirements

6.1 Database requirements

The database shall be designed using a relational model to support the Mentor Caring System.

It shall consist of multiple entities representing users, mentoring groups, records, and communication data, with clearly defined relationships.

The database shall include the following main entities:

- **User**

Attributes: **userID** (PK), **name**, **email**, **password**, **role**

Description: stores all system users including administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.

- **Student**

Attributes: **studentID** (PK), **major**, **status**

Relationship: linked to User entity

- **Mentor**
Attributes: `mentorID` (PK)
Relationship: linked to User entity
- **Group**
Attributes: `groupID` (PK), `academicYear`
Relationship: each group is assigned one mentor and multiple students
- **GroupMembership**
Attributes: `groupID` (FK), `studentID` (FK)
Description: represents many-to-many relationship between students and groups over different years
- **InterviewRecord**
Attributes: `recordID` (PK), `studentID` (FK), `mentorID` (FK), `date`, `content`
- **Appointment**
Attributes: `appointmentID` (PK), `studentID` (FK), `mentorID` (FK), `timeSlot`, `location`, `status`
- **Message**
Attributes: `messageID` (PK), `senderID` (FK), `receiverID` (FK), `content`, `timestamp`
- **Log**
Attributes: `logID` (PK), `userID` (FK), `loginTime`, `logoutTime`, `operation`

The database shall enforce the following relationships:

- One mentor can be assigned to multiple groups.
- One group can contain multiple students.
- One student can belong to different groups in different academic years.
- One mentor can create multiple interview records for different students.
- Messages shall support many-to-many communication between users.

The database shall also support:

- Role-based access control for different users
- Secure storage of user credentials
- Efficient retrieval of user and mentoring data
- Import of group allocation data from Excel files
- Preservation of historical mentoring records across academic years

6.2 Internationalization requirements

The system shall support internationalization to accommodate users from different linguistic backgrounds. English should be provided as the default language of the system interface. The system design should allow future extension to support additional languages if required.

6.3 Legal requirements

The system must comply with relevant institutional regulations regarding data privacy and information protection. Personal information of users, including student records and mentor information, must be protected and accessible only to authorized users.

6.4 Project reuse objective

The system should be designed with modular components so that certain modules (such as the user authentication module and database access module) can be reused in future university software systems if needed.

Appendix A: Glossary

MCP: Mentor Caring Program.

Faculty Consultant: Departmental advisor, responsible for designating the MCP coordinator and the import list.

MCP Coordinator: Departmental Coordinator, responsible for handling special cases forwarded by mentors.

Appendix B: Analysis Models

Context model in Section 2.1

Use case diagram in Section 2.2

Appendix C: Issues List

The following issues remain to be clarified during later stages of the system development process:

- Final confirmation of database schema
- Detailed access permissions for each user role
- Possible additional system features for future versions
- Final design of the user interface

These issues will be reviewed and resolved during the system design phase.